Witness camera: mechanical specification for CAD

Case, board sled, W-Mount interface, screw clamp and magnet puck · for Astra

Witness project · open hardware (CERN-OHL-P-2.0)
Spec rev A · October 2026

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1. Purpose and deliverables

Design a 3D-printable enclosure and modular mounting system for the Witness camera, a monitoring camera built on the FORIOT ESP32-S3-CAM board. Every part must print on a hobby FDM printer without supports unless noted, and must be fully parametric so the community can change it.

Deliverables, per part:

Format	Purpose
Parametric source (OpenSCAD or CadQuery preferred; FreeCAD acceptable)	Community edits. Every parameter in this document is a named variable.
STEP (AP214 or AP242)	Interchange with other CAD
STL, plus 3MF with print orientation already set	Printing
PNG renders: isometric, exploded, and each part in print orientation	Docs and README
`PARTS.md` listing each file, its material, orientation and hardware	Assembly

Units are millimeters. Tolerances not stated are ±0.2 mm on the printed part.

> **Dimensions marked TBV (to be verified)** come from vendor listings or clone boards, not from calipers. Before freezing geometry, measure them on a physical board. The Witness team will supply calipered

> values and photos on request. Every TBV value must be a top-level parameter.

2. System overview

The design has three tiers. Only the board kit changes when a different camera board is supported.

Tier	Parts	Changes per board?
Board kit	`sled`, `front_cover`	Yes
Universal body	`rear_shell`, `sd_door`, `button_pins`	No
Feet	`foot_clamp` (+ `clamp_screw`, `clamp_pad`, `clamp_nut`), `foot_magnet`, `foot_template`	No

The board kit and rear shell meet at a fixed **shell interface** (section 5.4). The shell and feet meet at the **W-Mount** (section 6). These two interfaces are the stable contracts. Publish them as separate parametric modules so others can design against them.

3. Reference board: FORIOT ESP32-S3-CAM

Orientation used throughout: hold the board component side toward you, with the **antenna end at the top** and the **two USB-C ports on the bottom edge**. “Front” is the component and camera side; “rear” is the side with the microSD slot and header pins.

Feature	Value	Status
PCB outline	62.6 × 28.3 mm	Vendor figure, TBV
PCB thickness	1.6 mm	TBV
Mounting holes	None. Retain by edges only.	TBV
Header rows	2 × 20 pins, 2.54 mm pitch, rows 25.4 mm apart	Vendor figure
Header pins on rear (if soldered)	Up to 8.5 mm below the PCB rear face	Design for both soldered and unsoldered
Tallest front component, excluding the camera	ESP32-S3-WROOM-1 shield, ~3.2 mm	TBV
Antenna keep-out	Top ~6 mm of the PCB (module antenna)	TBV
USB-C ports	Two, on the bottom edge. Label TTL (CH340) and OTG.	Centers TBV

Feature	Value	Status
USB-C plug overmold clearance	12.5 × 7.0 mm per plug, minimum	Typical cable
Buttons	RST and BOOT near the USB edge, front side	Positions TBV
LEDs	Power, TX, WS2812 RGB (GPIO 48), user LED (GPIO 2)	Positions TBV
microSD	Push-push slot on rear; card insertion edge	TBV
Camera	OV3660 module on a 24-pin 0.5 mm flex, lens on the front	Module size, lens Ø and flex length TBV

Camera module (TBV): typical modules are about 8.5 × 8.5 mm with an M8-class lens barrel about 8 mm in diameter. Wide-angle variants (120° and 160°) of the OV3660 fit the same connector. The front window must work with lens fields of view from 66° to 160° (section 5.2).

4. Design rules for all parts

Rule	Value
Target printers	0.4 mm nozzle, 0.2 mm layers, bed ≥ 180 × 180 mm
Materials	PETG (default) or ASA. PLA allowed for indoor prototypes. Optional TPU for pads.
Minimum wall	1.6 mm (four perimeters at 0.4 mm)
Sliding fit	0.20 mm clearance per side (`fit_slide`)
Snug/press fit	0.10 mm per side (`fit_press`)
Printed thread radial clearance	0.35 mm (`fit_thread`)
Overhangs	≤ 45° without supports; bridges ≤ 15 mm
Screws	M2 self-tapping into Ø1.7 pilot holes, or M2 heat-set inserts (Ø3.2 pocket) as an option (`use_inserts`)
Edge treatment	0.6 mm chamfer on outer edges; 1 mm fillets on hand-contact edges
Text	Debossed 0.4 mm, ≥ 4 mm tall: “WITNESS”, “TTL”, “OTG”, “SD”
Metal near antenna	No inserts, nuts or magnets within 15 mm of the antenna end

5. Body

5.1 Overall envelope

The interior is sized to accept the reference board and two planned future boards: the AI-Thinker ESP32-CAM (27 × 40.5 mm) and the Seeed XIAO ESP32S3 Sense (21 × 17.5 mm). Future sleds must fit without changing the rear shell.

Parameter	Default	Notes
`cavity_w` × `cavity_h`	32 × 67	Inside the shell, PCB plane
`front_clear`	9.0	PCB front face to inside of front cover, covers camera stack
`rear_clear`	9.5	PCB rear face to inside of rear shell, covers header pins plus SD
`wall`	2.0	Shell and cover walls
Approx. outside size	36 × 71 × 25	Width × height × depth, before the knuckle

`rear_clear` may be reduced to 4.5 in a “no headers” variant for users who desolder or never fit the header pins. Expose it as a parameter and ship both STLs.

5.2 Front cover (board kit)

- **Lens window.** A round aperture centered on the lens axis (`lens_x` , `lens_y` , TBV). The lens front sits no more than `lens_recess` = 1.0 mm behind the cover’s outer face. Aperture diameter = `lens_od + 2 × lens_recess × tan(fov_half) + 0.6` , with `fov_half` = 80° by default so 160° lenses do not vignette. Add a 45° outward chamfer.
- **Camera retention.** A square pocket on the inside, sized to the camera module plus `fit_slide` , holds the module square to the window. A printed spring finger or a 1 mm foam pad behind the module presses it forward. The lens must not touch the window edge.
- **Flex routing.** A channel from the camera pocket to the board’s FPC connector, minimum bend radius 1.5 mm, with no sharp edges along the path. The flex must not be creased.
- **Light pipes.** Optional 2 mm holes over the power LED and the WS2812 (`led_pipe = true`). Keep them at least 15 mm from the lens axis and add a 0.8 mm baffle ring so LED light cannot leak into the lens.
- **Button access.** Ø2.5 mm holes over RST and BOOT for a paperclip, or captive printed push pins (`button_pins` part) with a 0.4 mm flexure retention lip.
- **Closure.** Mates to the rear shell through the shell interface (5.4).

5.3 Sled (board kit)

- Holds the PCB by its long edges. Two rails with slots `pcb_t + 0.15` wide and 1.2 mm deep. The board slides in from the USB end toward an end stop at the antenna end.
- A flexing detent tab (0.8 mm thick, 6 mm long) clicks over the USB edge of the PCB to stop it sliding back out.
- The sled keeps the PCB centered in the cavity so the USB-C ports line up with the rear-shell openings to ±0.3 mm. Register it to the shell with two locating pins (Ø2.0) at fixed positions defined by the shell interface.
- Underside cutouts leave clear access to the SD slot and to header pins.
- Thermal: no material within 1 mm of the module shield's top face; leave an air gap to the vents.

5.4 Shell interface (fixed contract)

This is the boundary between the board kit and the universal rear shell. It must stay fixed so any future board kit fits any rear shell.

Feature	Spec
Joint	Tongue on the front cover (1.2 wide × 1.5 tall), groove on the rear shell (`1.2 + 2 × fit_slide`) around the full perimeter
Fasteners	2 × M2 × 8 screws through the front cover into rear-shell bosses, on the long sides, at least 15 mm from the antenna end
Sled locating	2 × Ø2.0 pins on the rear-shell floor; sled holes Ø2.0 + `fit_press`
Datum	Interior cavity center, PCB plane at `rear_clear` above the rear-shell floor

5.5 Rear shell (universal)

- **USB-C openings.** Two openings on the bottom face, each sized to the plug overmold (12.5 × 7.0 mm plus `fit_slide`), centered on the port positions. Deboss "TTL" and "OTG" beside them. If the two port centers are too close for two full overmold openings, use one combined slot and note it.
- **Cable strain relief.** A zip-tie bridge (4 × 2 mm slot pair) on the bottom face beside the USB openings.
- **SD door.** A sliding or snap door (`sd_door` part) over the card slot. The card must be insertable and ejectable (push-push) with the case closed and the door open.
- **Vents.** Slots 1.2 mm wide on both long sides, alongside the ESP32 module and away from the lens. Total open area ≥ 60 mm². No direct line of sight from a vent to the camera's inside face, to keep out stray light.
- **W-Mount knuckle.** A two-finger knuckle on the rear face (section 6), centered horizontally and placed 22 mm from the USB end (`knuckle_y`). The tilt axis is parallel to the short edge, so the camera tilts up and down.

- **Antenna end.** Keep the top 15 mm of the shell free of metal and of solid walls thicker than `wall`.

6. W-Mount interface (fixed contract)

The W-Mount follows the de facto GoPro finger mount, so off-the-shelf GoPro accessories work. **Astra must check the final parameters against a genuine GoPro buckle and adjust them until both mate with a firm, wobble-free fit.**

Parameter	Default	Notes
`finger_t`	3.0	Finger thickness
`finger_gap`	3.2	Gap between fingers
`finger_r`	7.5	Outer radius around the pivot
`finger_reach`	9.0	Pivot center to base surface
`bolt_d`	5.3	Clearance for M5
Camera side (`knuckle2`)	2 fingers	Total width 9.2
Foot side (`knuckle3`)	3 fingers	Total width 15.4. One outer finger has a captive M5 hex-nut pocket, 8.1 mm across flats × 4.0 deep.
Bolt	M5 × 25 thumbscrew or socket-head bolt	Printed knob cap `m5_knob` for socket heads

Print fingers so the layers stack across the finger thickness (finger faces parallel to the bed). This avoids layer splits at the pivot. Fillet the finger roots with r = 1.5.

Publish `knuckle2()` and `knuckle3()` as standalone modules in their own file (`w_mount.scad` or `w_mount.py`), with a doc comment showing how to attach them to a custom base.

7. Feet

7.1 Screw clamp (`foot_clamp`)

A C-clamp for shelf edges, robot frames, 2020/3030 extrusion, bench tops and tubes.

Parameter	Default	Notes
`jaw_open`	35	Maximum opening; 0 is the minimum
`throat`	25	Depth from jaw edge to the frame's back
`frame_section`	10 × 14	Cross-section of the C frame. Stiff enough in PETG for hand tightening.

Parameter	Default	Notes
Thread	Tr14 × 3 trapezoidal, printed	`fit_thread` clearance
Screw length	`jaw_open + 20`	
Knob	Ø30 knurled, or a 40 mm T-bar (`knob_style`)	
Pad	Ø20 swivel pad, snap-fit ball (Ø8) into a socket on the screw tip	TPU or PETG face, 1 mm grip ridges
Fixed jaw	20 × 14 face with 1 mm ridges; optional TPU overlay	
Knuckle	`knuckle3` on the back of the frame, axis parallel to the jaw faces by default	`knuckle_axis` = 0° or 90°

Print orientation. Print the C frame lying flat, with the C shape in the XY plane, so bending loads run along the layers. The internal thread would then print sideways, so the frame instead has a hex pocket that accepts a separate printed **`clamp_nut`**, printed upright for clean threads and pressed in (`fit_press`). Provide a variant pocket that accepts a steel M8 nut, with an M8 bolt in place of the printed screw (`steel_screw = true`). Print the screw upright.

Load check. The clamp must hold a camera (under 60 g) cantilevered 60 mm from the jaw under hand tightening without slipping, and survive 20 tighten and loosen cycles without the thread wearing out.

7.2 Magnet puck (`foot_magnet`)

A puck that sticks to steel enclosures, machine frames and steel plates. Magnets slide in from the side and are glued in place.

Parameter	Default	Notes
`puck_d` × `puck_h`	40 × 9	
`magnet_shape`	`disc`	`disc` or `block`
Disc magnets	4 × Ø10 × 3 mm N52	Pockets `Ø10.2 × 3.15`
Block magnets (alt.)	2 × 20 × 10 × 3 mm N52	Pockets `20.2 × 10.2 × 3.15`
`skin`	0.4	Plastic between the magnet and the contact face. 0 leaves the magnet face flush and exposed.
Pocket entry	Each pocket opens to the puck's outer edge through a slot of the same section, so a magnet slides in radially	

Parameter	Default	Notes
Retention	A 0.15 mm detent bump at the slot mouth, plus a Ø1.8 glue port from the top face into each pocket for CA or epoxy	
Contact face	Optional 0.8 mm recess for a 40 mm self-adhesive silicone or TPU anti-slip disc	`grip_recess`
Knuckle	`knuckle3` on the top face, centered	
Pry notch	6 × 2 mm notch on the rim to lever the puck off steel	

Print orientation. Print the contact face down on the bed so the skin is smooth and flat. Pocket ceilings are bridges of 10–20 mm, which is acceptable.

Notes for the user guide. Magnet polarity does not matter on thick steel. Alternating N/S between pockets grips thin sheet better. Four Ø10 × 3 N52 discs behind a 0.4 mm skin hold the camera on a vertical steel face with a large safety margin. Keep the puck at least 15 mm from the camera's antenna end.

7.3 Foot template (`foot_template`)

A minimal example of a custom foot: a 30 × 30 × 4 base plate with `knuckle3`, two M3 through holes and a VHB tape recess. Include a short comment block that explains how to attach `knuckle3()` to any geometry. This is the starting point for community feet such as zip-tie, 2020 T-slot, 1/4"-20 tripod adapter and suction cup.

8. Assembly (for `PARTS.md`)

1. Insert the microSD card into the board.
2. Seat the camera module in the front-cover pocket and route the flex along its channel.
3. Slide the PCB into the sled until the detent clicks.
4. Drop the sled onto the rear-shell locating pins. Check the USB-C ports line up with their openings.
5. Close the front cover onto the rear shell and fit 2 × M2 × 8 screws.
6. Press the nut into the foot's `knuckle3`, join it to the camera's `knuckle2`, and fit the M5 bolt.

Hardware list: 2 × M2 × 8 self-tapping screws; 1 × M5 × 25 bolt or thumbscrew; 1 × M5 nut; magnets per 7.2; optional M8 × 60 bolt and nut for the steel clamp variant; optional TPU pads.

9. Acceptance checks

#	Check	Pass criterion
1	Board fit	Board slides in and out by hand. No PCB flex. USB-C plugs seat fully in both ports.
2	Optics	No vignetting at 160° FOV, no LED glare in a dark-room capture, lens square to the window within 2°
3	SD access	Card inserts and ejects with the case closed and the door open
4	Buttons	RST and BOOT can be pressed with the case closed
5	Thermal	After 2 hours recording at 22 °C ambient, shell surface stays under 45 °C and the camera keeps running
6	W-Mount	Mates with a genuine GoPro buckle and with the printed feet; holds a set tilt with the bolt hand-tight
7	Clamp	Load check in 7.1
8	Magnet	Holds on a vertical steel plate at the 60 mm cantilever; a 0.4 mm skin survives 50 attach and detach cycles
9	Printability	Every part prints without supports in PETG at 0.2 mm, in the 3MF orientation
10	Wi-Fi	RSSI at 5 m is no worse than 6 dB below the bare board, measured with the magnet puck on a steel plate

10. Open questions for Astra

1. Confirm the TBV dimensions in section 3 from a physical board, especially the lens position, USB-C centers, button and LED positions, and SD insertion edge.
2. If a camera flex is too short to put the lens where the front cover needs it, propose the lens position and we will standardize on it.
3. Advise whether OpenSCAD or CadQuery gives the community the better parametric source for these shapes.